

# Notes: Huang and Sacchetto (JFE, 2026)

## Bonding with Risk: Corporate Investment and Savings in Risky Financial Assets

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## 1 Big picture

- **Question.** Why do non-financial firms hold risky financial assets, and can this behavior be explained by rational precautionary motives rather than only by agency problems, taxes, or behavioral biases?
- **Setting.** The paper combines a dynamic model of corporate investment and cash management with a new firm-level dataset on financial asset holdings constructed from SEC 10-K filings for US firms from 2009 to 2018.
- **Core challenge.** In standard dynamic cash-holding models, firms transfer liquidity only through safe assets. But in the data many firms also hold risky securities, especially bonds. The authors ask whether risky assets can provide *state-contingent liquidity*: high payoffs exactly in states when firms need funding for investment.
- **Main methodological idea.** The paper extends dynamic investment models by allowing firms to allocate precautionary savings between a safe one-period asset and a risky bond. The key sufficient statistic is the **financing deficit beta**, i.e. the sensitivity of the firm’s financing deficit to the return on the risky asset.
- **Main theoretical result.** Firms hold more risky financial assets when their financing deficits covary more positively with risky bond returns. If bond returns are high precisely when investment needs are high and internal funds are tight, risky bonds hedge costly external financing.
- **Main empirical result.** Using a sample of **19,367 firm-year observations** for **2,886 US firms**, the paper finds that risky financial asset holdings are positively related to financing

deficit betas based on bond-market returns. The link is strongest for betas tied to **medium- and long-term interest-rate risk**.

- **Main quantitative result.** In the baseline cross-sectional regressions, a one-standard-deviation increase in the financing deficit beta is associated with about **0.81% to 0.88% more risky financial assets relative to total assets**, and about **1.74% to 2.02% more risky assets as a share of total financial assets**.
- **Heterogeneity.** Precautionary motives are especially strong for **small firms, high-volatility firms**, and **R&D-intensive firms**.
- **Counterfactual result.** In the model, shutting down access to risky financial assets lowers firm value by about **0.35%**, reduces investment and profits, and increases both the amount and frequency of external financing.
- **Economic takeaway.** Risky financial assets can be a rational liquidity-management tool. They are valuable because they move liquidity toward states in which firms' financing deficits are large, reducing the need for costly issuance and supporting real investment.

## 2 Data and measurement (key objects)

### 2.1 Observed data sources and sample construction

- **Main accounting sample.** The starting point is Compustat annual data from **2009–2018**, beginning when SFAS No. 157 became implemented.
- **Main sample restrictions.** The paper drops:
  - regulated utilities (SIC 4900–4999),
  - financial firms (SIC 6000–6999),
  - observations with missing or negative total assets or property, plant, and equipment,
  - observations with total assets or PPE below \$5 million,
  - observations where measured risky financial assets exceed total assets, because these are likely data-extraction mistakes.
- **SEC filing data.** The novel part of the data is the collection of fair-value information on financial assets from the notes to 10-K filings in EDGAR, matched to firms using CIK and fiscal year.
- **Additional source.** The authors also complement the sample with extra firm-year observations from Darmouni and Mota (2024).
- **Final sample.** The final empirical sample contains **19,367 firm-year observations** for **2,886 firms**.

## 2.2 How the paper measures safe and risky financial assets

- **Safe assets.** The classification is built around the types of securities in Compustat items CH and IVST. Safe assets include cash, bank receivables, bank drafts, bank acceptances, deposits, checks, letters of credit, money orders, commercial paper, Treasuries, and money market funds.
- **Risky assets.** Risky financial assets are all remaining disclosed securities that are *not* on the safe-assets list, after excluding restricted assets, pension assets, liabilities, assets held for compensation, and hedging-related positions.
- **Why the algorithm is built this way.**
  - Omitted items in 10-K disclosure tables are more likely to be safe assets.
  - Safe asset names are relatively standardized across firms.
  - So the paper first detects the safe items and then treats the remaining securities as risky.
- **Machine-learning procedure.** The classification algorithm is trained on **1,500** 10-K filings, containing **527** target tables, of which **333** are manually labeled as containing relevant information on corporate financial assets. The model uses an  $n$ -gram representation and an L1-regularized logistic regression classifier.
- **Out-of-sample accuracy.** On a random test sample of **500** 10-K filings containing **112** target tables, the algorithm achieves an overall accuracy rate of about **83.93%**.

## 2.3 Main variables

- **Risky:**

$$\text{Risky}_{jt} = \frac{\text{fair value of risky financial assets}}{\text{total assets}}.$$

- **Safe:**

$$\text{Safe}_{jt} = \frac{\text{safe financial assets}}{\text{total assets}},$$

where the empirical proxy comes from Compustat cash-like holdings.

- **Financial-portfolio composition:**

$$\text{FinComp}_{jt} = \frac{\text{risky financial assets}}{\text{total financial assets}}.$$

- **Financing deficit beta.** This is the paper’s key empirical measure of hedging demand. It is estimated at the firm level from time-series regressions of the firm’s quarterly financing deficit on a risky asset return.
- **Two main beta measures.**
  - **BetaAgg:** based on the return on the Bloomberg US Aggregate Bond Market Index.
  - **BetaTre:** based on the reverse change in the 10-year Treasury yield.

- **Residual volatility control.** The paper also computes the standard deviation of the residuals from the first-stage beta regression, called **FDResiVol**, to separate hedging incentives from the general level of financing-deficit uncertainty.
- **Other controls.** Lagged Tobin’s  $Q$ , lagged profitability, and lagged firm size are included in the main cross-sectional regressions.

## 2.4 Key descriptive facts

- From Table 3, the firm-year means are:
  - **Risky** = **0.042**,
  - **Safe** = **0.143**,
  - **FinComp** = **0.151**,
  - lagged  $Q$  = **1.884**,
  - lagged profitability = **0.076**,
  - lagged size = **6.680**.
- The financing-deficit-beta measures are economically meaningful and dispersed:
  - **BetaAgg**: mean **0.409**, standard deviation **3.208**,
  - **BetaTre**: mean **1.010**, standard deviation **12.255**.
- The corresponding residual-volatility measures are also large on average, with means of about **27.6** to **27.7**.
- S&P 500 firms hold much more risky financial assets than non-S&P 500 firms. By 2017, risky financial assets of S&P 500 firms exceed **10% of total assets** and **50% of total financial assets**. Non-S&P 500 firms hold comparable amounts of safe assets, but much smaller — though still non-negligible — risky portfolios.
- High- $Q$ , low-profitability, and high-tech firms hold more financial assets, both safe and risky, than their comparison groups.

## 3 Models and empirical specifications (all equations + interpretation)

### 3.1 Technology, shocks, and capital accumulation

The firm’s operating profits are

$$\pi_{jt} = \exp(x_t + z_{jt})k_{jt}^\alpha,$$

where  $x_t$  is an aggregate profitability shock and  $z_{jt}$  is an idiosyncratic profitability shock.

Both shocks follow AR(1) processes:

$$x_t = \rho_x x_{t-1} + \sigma_x \varepsilon_t^x,$$

$$z_{jt} = \rho_z z_{jt-1} + \sigma_z \varepsilon_{jt}^z.$$

Capital evolves according to

$$k_{j,t+1} = (1 - \delta)k_{jt} + i_{j,t+1},$$

and the firm pays quadratic capital adjustment costs

$$\text{Adj}_{jt}^k = \frac{\psi_i}{2} \left( \frac{k_{j,t+1} - (1 - \delta)k_{jt}}{k_{jt}} \right)^2 k_{jt}.$$

- **Interpretation.** The firm has persistent profitability and costly capital adjustment. These two ingredients create a dynamic motive to save: a firm wants liquidity today because tomorrow's investment needs depend on persistent shocks.

### 3.2 Stochastic discount factor and financial securities

The stochastic discount factor is

$$\log M(x_t, x_{t+1}) = \log(\eta) - \gamma(x_{t+1} - x_t),$$

which implies a state-dependent risk-free rate

$$r_f(x_t) = \frac{1}{\mathbb{E}[M(x_t, x_{t+1}) \mid x_t]} - 1.$$

The firm can save in two assets:

- a safe one-period security,
- a risky bond.

The safe asset price is

$$q_c(x_t) = \mathbb{E}[M(x_t, x_{t+1})(1 + r_f)] = \frac{1 + r_f}{1 + r_f(x_t)}.$$

The risky bond has coupon  $r_f$ , maturity  $1/\mu$ , and price

$$q_s(x_t) = \mathbb{E}\left[M(x_t, x_{t+1})(\mu + r_f + (1 - \mu)q_s(x_{t+1})) \mid x_t\right].$$

- **Interpretation.** The risky bond is risky because part of its payoff comes from the resale price  $q_s(x_{t+1})$ , which moves with the aggregate state. The maturity parameter  $\mu$  governs how much interest-rate risk the bond carries.
- **Why a bond?** The paper argues that this matches the data, because most corporate risky financial assets are bond-like securities. It is also computationally convenient relative to modeling a generic risky equity claim.

### 3.3 Cash flows, financing frictions, and the Bellman problem

If the firm enters period  $t$  with safe holdings  $c_{jt}$  and risky bond holdings  $s_{jt}$ , then the current value of the financial portfolio is

$$(1 + r_f)c_{jt} + (\mu + r_f)s_{jt} + q_s(x_t)(1 - \mu)s_{jt}.$$

Taxable income is

$$TI_{jt} = \exp(x_t + z_{jt})k_{jt}^\alpha - \delta k_{jt} + r_f c_{jt} + r_f s_{jt}.$$

Bond trading costs are linear:

$$\text{Adj}_{jt}^s = \psi_s |s_{j,t+1} - (1 - \mu)s_{jt}|.$$

Gross cash flow is

$$\begin{aligned} e_{jt} = & (1 - \tau)TI_{jt} + \delta k_{jt} + c_{jt} + \mu s_{jt} + q_s(x_t)(1 - \mu)s_{jt} \\ & - i_{j,t+1} - \text{Adj}_{jt}^k - q_c(x_t)c_{j,t+1} - q_s(x_t)s_{j,t+1} - \text{Adj}_{jt}^s. \end{aligned}$$

Net distributions are

$$d_{jt} = e_{jt} - \Lambda_{jt},$$

with external financing costs

$$\Lambda_{jt} = \xi |e_{jt}| \mathbf{1}\{e_{jt} < 0\}.$$

The Bellman equation is

$$v(x, z, s, c, k) = \max_{s', c', k'} \{d(x, z, s, c, k, s', c', k') + \mathbb{E} [M(x, x')v(x', z', s', c', k') \mid x, z]\}.$$

- **Interpretation.** The state vector is  $(x, z, s, c, k)$  and the firm jointly chooses next-period risky assets, safe assets, and physical capital. The external financing wedge  $\xi$  is the fundamental reason liquidity matters in the model.

### 3.4 Precautionary motive and the financing deficit beta in the model

The paper's core mechanism is easiest to see through the financing-deficit condition. Define the financing deficit as the difference between desired physical investment spending and internally generated operating resources. In the model,

$$FD = \frac{k' - (1 - \delta)k + \text{Adj}^k - (1 - \tau)\exp(x + z)k^\alpha - \tau\delta k}{k}.$$

The firm incurs costly external finance when this financing deficit exceeds the cash flow coming from its financial assets.

A one-dollar shift from safe assets into the risky bond changes next-period available funds by

$$(1 - \mu)(q_s(x) - 1).$$

- If  $q_s(x) > 1$ , the risky bond delivers more cash than the safe asset in that state.
- If  $q_s(x) < 1$ , it delivers less.

Because the estimated risky bond has a payoff structure that is better in good aggregate states, and because the firm's financing deficit tends to be larger when aggregate profitability is high, the risky bond transfers liquidity toward precisely the states in which the firm wants to invest more.

The paper summarizes this hedge motive with the **financing deficit beta**:

$$FD_{j,t+1} = \alpha_{jt} + \beta_{jt}^D R_{t+1}^B + \varepsilon_{j,t+1},$$

where  $R_{t+1}^B$  is the risky bond return. In model terms,

$$\beta_{jt}^D = \frac{\text{Cov}_t(FD_{j,t+1}, R_{t+1}^B \mid x_t, z_{jt}, s_{jt}, c_{jt}, k_{jt})}{\text{Var}_t(R_{t+1}^B \mid x_t)}.$$

- **Interpretation.**  $\beta^D$  is the paper's sufficient statistic for hedging demand. A firm with higher  $\beta^D$  has a financing deficit that covaries more positively with risky-bond returns, so risky bonds are more useful as a hedge.

### 3.5 Structural estimation

- The model has **13 parameters**. Seven are calibrated or pinned down from prior literature and macro moments; six are estimated by SMM.
- **Calibrated parameters.** The paper sets:
  - $\rho_x = 0.8140$ ,
  - $\sigma_x = 0.0073$ ,
  - $\gamma = 15.6$ ,
  - $\eta = 0.96$ ,
  - $\mu = 0.4$ ,
  - $\tau = 0.14$ ,
  - $\delta = 0.10$ .
- **Estimated by SMM.** The estimated parameters are:

$$(\rho_z, \sigma_z, \alpha, \psi_i, \xi, \psi_s).$$

- **Target moments.** The SMM objective matches 11 moments:
  - average risky financial assets,
  - average safe financial assets,
  - average investment,
  - average  $Q$ ,



- average profitability,
- median distributions,
- variance of profitability,
- variance of distributions,
- autocorrelation of profitability,
- autocorrelation of investment,
- autocorrelation of distributions.

- **Simulation design.** The model is simulated in **50 samples of 3,000 firms over 50 years.**

The baseline parameter estimates are:

$$\begin{aligned}\hat{\rho}_z &= 0.7576, & \hat{\sigma}_z &= 0.3546, \\ \hat{\alpha} &= 0.8552, & \hat{\psi}_i &= 0.4756, \\ \hat{\xi} &= 0.0951, & \hat{\psi}_s &= 5.7723 \text{ bps.}\end{aligned}$$

- **Interpretation.**

- $\hat{\xi} = 0.0951$  means external financing costs are about **9.51% of proceeds raised**.
- $\hat{\psi}_s \approx 5.77$  basis points implies modest but nontrivial trading costs for risky bonds.
- The model matches especially well the means of risky assets, safe assets, investment, and Tobin's  $Q$ .

### 3.6 Policy functions and simulated cross-sectional predictions

- **Policy functions (Figure 3).**

- When aggregate profitability  $x$  is high, the firm liquidates both risky and safe assets to finance investment and avoid external financing.
- When  $x$  is low, the firm cuts investment, distributes more to investors, and prefers not to trade the risky bond much because of trading costs.
- At median  $x$ , the firm largely rolls over financial assets and uses internal funds to invest while keeping distributions close to zero.

- **Mechanism across firm size (Figure 4).** Small firms have financing deficits that increase more sharply with aggregate conditions than large firms, so the hedge value of risky bonds is greater for them.
- **Simulated reduced-form prediction (Table 2).** In the simulated data, a one-standard-deviation increase in the financing deficit beta raises risky financial asset holdings by about
  - **0.45% of total assets**, and
  - **4.81% of total financial assets**.

The beta coefficient is larger for small firms than for large firms.

### 3.7 Empirical specifications

The first-stage empirical beta regression is

$$FD_{jt} = \alpha_j + \beta_j^D R_{t-1}^B + \varepsilon_{jt},$$

where  $FD_{jt}$  is the quarterly financing deficit of firm  $j$  and  $R_{t-1}^B$  is either a bond-index return or the reverse change in a Treasury yield.

The main second-stage cross-sectional regression is

$$FinAssets_{jt} = \zeta_{sic,t} + \phi \hat{\beta}_j^D + Controls_{j,t-1} + \varepsilon_{jt},$$

where  $FinAssets_{jt}$  is either **Risky**, **Safe**, or **FinComp**, and the controls include lagged  $Q$ , profitability, size, and financing-deficit residual volatility.

The paper also studies whether risky financial assets predict real policies using

$$Y_{jt} = \phi_1 Risky_{j,t-1} + \phi_2 Safe_{j,t-1} + Controls_{j,t-1} + \alpha_j + \zeta_t + \varepsilon_{jt},$$

where  $Y_{jt}$  is either the investment rate or external financing.

- **Interpretation.** The empirical design mirrors the model exactly: estimate each firm's hedge demand in stage 1, then ask in stage 2 whether firms with stronger hedge demand actually hold more risky financial assets.

### 3.8 Counterfactual design

The main counterfactual experiment solves the model under the restriction

$$s = s' = 0,$$

so firms are not allowed to invest in risky financial assets.

The paper compares this economy to the baseline one along the following margins:

- capital,
- investment,
- profits,
- safe and risky savings,
- issuance amounts and issuance frequency,
- firm value.

It also studies comparative statics in the risky-asset trading-cost parameter  $\psi_s$ .

## 4 Identification and instruments (what makes the estimates credible)

### 4.1 What identifies the structural parameters

- **Technology-shock parameters.** The persistence and volatility of the idiosyncratic shock,  $(\rho_z, \sigma_z)$ , are identified from profitability dynamics and, in subsample exercises, from firm-level TFP autocorrelation regressions.
- **Returns-to-scale parameter  $\alpha$ .** Average Tobin's  $Q$  is highly informative about  $\alpha$ .
- **Capital adjustment cost  $\psi_i$ .** Average investment is especially sensitive to  $\psi_i$ .
- **External financing cost  $\xi$ .** Higher  $\xi$  increases the value of precautionary savings, so it raises both safe and risky asset holdings.
- **Risky-asset trading cost  $\psi_s$ .** Higher  $\psi_s$  pushes firms away from risky assets and toward safe assets. So the *composition* of financial assets helps separately identify  $\psi_s$  from  $\xi$ .
- **Why Figure 2 matters.** The comparative statics show that these parameters move different target moments in different ways, which is the paper's practical identification argument.

### 4.2 What identifies the financing-deficit-beta effect in the data

- **Stage 1: within-firm time series.** For each firm, the paper estimates how the quarterly financing deficit co-moves with bond-market returns. This yields a firm-specific financing deficit beta.
- **Stage 2: cross-section.** The paper asks whether firms with larger estimated betas hold more risky financial assets in the cross-section.
- **Controls that matter.**
  - **FDResiVol** controls for the overall volatility of financing deficits that is *not* explained by bond returns.
  - Lagged  $Q$ , profitability, and size absorb systematic differences in investment opportunities, internal funds, and scale.
  - Industry-year fixed effects absorb broad industry and time shocks.
- **Interpretation.** The paper is not simply testing whether risky firms hold risky assets. It is testing whether firms whose *funding gaps line up with bond returns* hold more risky assets, conditional on general volatility and other firm characteristics.

### 4.3 Why the financing deficit beta is a credible hedge-demand proxy

- The model predicts that the beta should be strongly related to **Risky** and **FinComp**, but only weakly related to **Safe**. That is exactly what the data show in Table 4.
- Betas based on longer-maturity Treasury instruments matter more than betas based on very short rates. This pattern is hard to reconcile with a generic “riskiness” story and instead points specifically to hedging of medium- and long-term interest-rate risk.

- When alternative risk factors are included, the Treasury-based beta remains robust. This reinforces the interpretation that interest-rate risk is the central channel.

#### 4.4 Data credibility and robustness

- The algorithmic extraction of risky financial assets is tested out of sample and achieves about 84% accuracy.
- The paper adjusts second-stage standard errors using Murphy–Topel corrections because the financing deficit beta is a generated regressor.
- The main results are robust to:
  - rolling-window, time-varying betas,
  - alternative volatility controls,
  - stronger winsorization and percentile-based beta transforms,
  - Compustat IVST-based proxies,
  - financing-deficit definitions using intangible capital.

#### 4.5 Relation to the literature

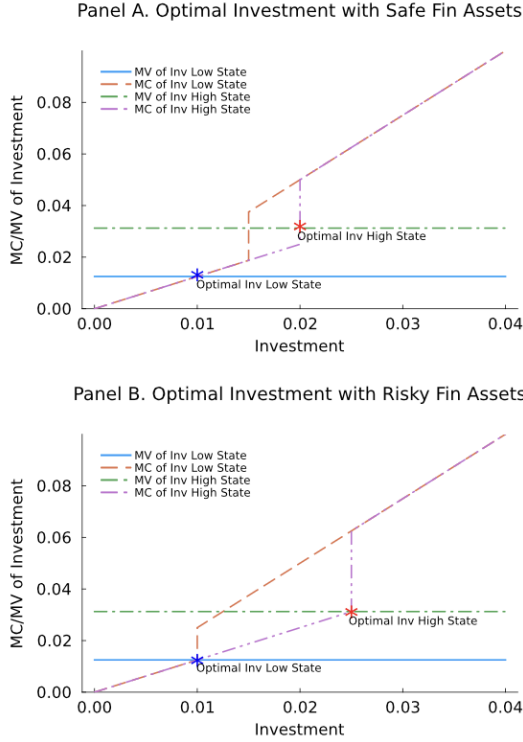
- **Corporate savings.** The paper extends dynamic cash-holding models such as Riddick and Whited (2009), Nikolov and Whited (2014), and Begenau and Palazzo (2021) by allowing firms to save in both safe and risky securities.
- **Governance and tax views of risky asset holdings.** It complements Duchin et al. (2017) and Darmouni and Mota (2024), who emphasize agency, overconfidence, and taxation, by highlighting a rational hedging motive.
- **Risk management.** The mechanism links naturally to the Smith–Stulz and Froot–Scharfstein–Stein view that firms hedge in order to reduce the costs of external financing and support investment.

### 5 Tables and figures

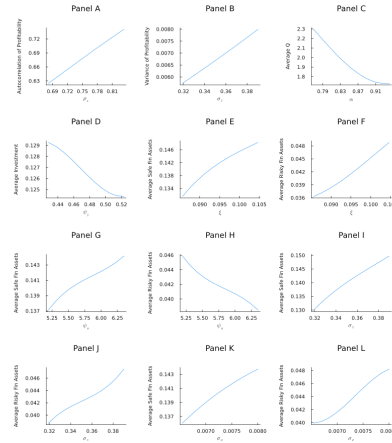
#### 5.1 Figure 1 and Figure 2: economic mechanism and comparative statics

**Read:**

- Figure 1 is the cleanest picture of the model’s intuition. The risky bond shifts internal funds from low- $x$  states to high- $x$  states, so it relaxes financing constraints when investment opportunities are best.
- The figure explains why the paper focuses on the financing deficit rather than simply on cash-flow volatility: the relevant object is whether liquidity arrives in the *same states* in which the firm wants to invest.



**Fig. 1.** Precautionary motives and risky financial assets. This figure illustrates the first-order conditions for optimal investment to highlight the precautionary role of risky financial assets in the model. Both panels plot the marginal value and marginal cost of investment, defined in Eq. (17) in the simple case of  $\alpha = 1$ , for high and low states of the aggregate profitability shock,  $x$ . In Panel A, the firm saves in the risk-free security, while in Panel B the firm invests in the risky financial asset. Panel A illustrates a scenario in which optimal investment is capped by external financing frictions when  $x$  is high, but the firm has extra liquidity when  $x$  is low. In Panel B, by diverting its savings from the safe to the risky financial asset, the firm can alleviate the financing constraints in the high state, without increasing the total amount of savings.



**Fig. 2.** Comparative statics. This figure presents comparative statics showing how key moments respond to changes in underlying structural parameters. For each plot, we set the parameters to the values in Table 1, and only vary the relevant parameter. Panels A, B, C, and D display the autocorrelation of profitability, the variance of profitability, average Tobin's  $Q$ , and the average investment rate, respectively, against the persistence of the idiosyncratic profitability shock,  $\rho_p$ . Panels E through L plot the average safe and risky financial assets, respectively, against the external financing cost parameter,  $z$ , in Panels E and F, the trading cost of risky financial assets,  $w_r$ , in Panels G and H, the standard deviation of the conditional idiosyncratic profitability shock,  $\sigma_p$ , in Panels I and J, and the conditional standard deviation of the aggregate profitability shock,  $\sigma_x$ , in Panels K and L. Moment definitions are in Table A.3.

- Figure 2 provides the identification map for the structural estimation. Higher external financing costs raise both safe and risky savings; higher risky-asset trading costs rotate the portfolio away from risky assets toward safe assets; higher volatility raises precautionary saving overall.
- This is why the model can separately identify  $\xi$  and  $\psi_s$ : the two parameters move the level and composition of savings differently.

## 5.2 Table 1, Figure 3, and Figure 4: baseline structural estimates and policy functions

Table 12

Risky financial assets and firm value. This table reports the results of a counterfactual experiment based on the model to evaluate the effects and value of the option to invest in risky financial assets. We compare the full model to a counterfactual version in which firms are restricted from holding risky financial assets. Both models are solved using the estimated parameters reported in Table 1. We simulate two samples of firms from the full and counterfactual models and compare their outcomes across operating performance, financing behavior, and firm value. Column 1 reports the mean differences in each outcome, scaled by average capital in the full model—except for issuance frequency and firm value, which are scaled by their respective means. Columns 2 to 5 report analogous differences for subsamples defined by business cycle conditions and firm size, using the same scaling relative to conditional means. Boom and bust periods are defined by positive and negative aggregate profitability shocks, respectively; below and above median size firms are defined relative to the yearly median of total assets.

|                                  | % Differences No Risky vs Full Model |                     |                     |                      |                      |
|----------------------------------|--------------------------------------|---------------------|---------------------|----------------------|----------------------|
|                                  | (1)<br>Full Sample                   | (2)<br>Boom Periods | (3)<br>Bust Periods | (4)<br>< Median Size | (5)<br>> Median Size |
| Capital                          | -0.1685                              | -0.1617             | -0.1233             | -0.2289              | -0.1501              |
| Investment                       | -0.0165                              | -0.0170             | 0.0150              | -0.0398              | -0.0093              |
| Profit                           | -0.0250                              | -0.0213             | -0.0210             | -0.0305              | -0.0233              |
| Financial Assets                 | -0.3566                              | -0.2282             | -0.3305             | -0.6704              | -0.2610              |
| Tax Costs on Financial Assets    | -0.0018                              | -0.0011             | -0.0016             | -0.0033              | -0.0013              |
| Trading Costs                    | -0.0008                              | -0.0006             | -0.0006             | -0.0016              | -0.0006              |
| Cash Flows from Financial Assets | -0.0110                              | -0.0385             | -0.0824             | -0.0207              | -0.0080              |
| Distributions                    | -0.0176                              | -0.0439             | -0.1156             | -0.0086              | -0.0203              |
| Issuance Amount                  | 0.0471                               | 0.0879              | 0.0008              | 0.0997               | 0.0311               |
| Issuance Costs                   | 0.0045                               | 0.0084              | 0.0001              | 0.0095               | 0.0030               |
| Issuance Frequency               | 4.6964                               | 1.2243              | 19.2459             | 5.0150               | 4.2484               |
| Firm Value                       | -0.3499                              | -0.3089             | -0.2143             | -0.4594              | -0.3008              |

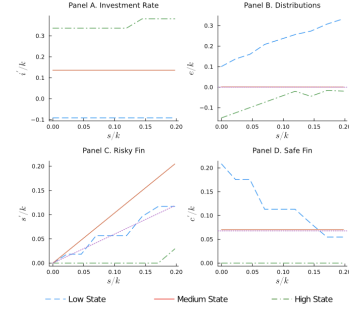


Fig. 3. Policy functions. This figure plots the following optimal policy functions for different levels of the aggregate profitability shock  $x$ : the investment rate ( $i/k$ ) in Panel A, distributions/external financing ( $d/k$ ) in Panel B, risky financial asset holdings ( $r/k$ ) in Panel C, and safe financial asset holdings ( $s/k$ ) in Panel D against current-period risky financial asset holdings ( $r/k$ ). The dotted reference lines in Panels B, C, and D represent the loci of points corresponding to zero distributions ( $d/k = 0$ ), zero trading in risky financial assets ( $r/k = (1 - \mu)r/k$ ), and zero adjustment in safe financial assets ( $s/k = c/k$ ), respectively. The optimal policies are based on the estimated parameters in Table 1.

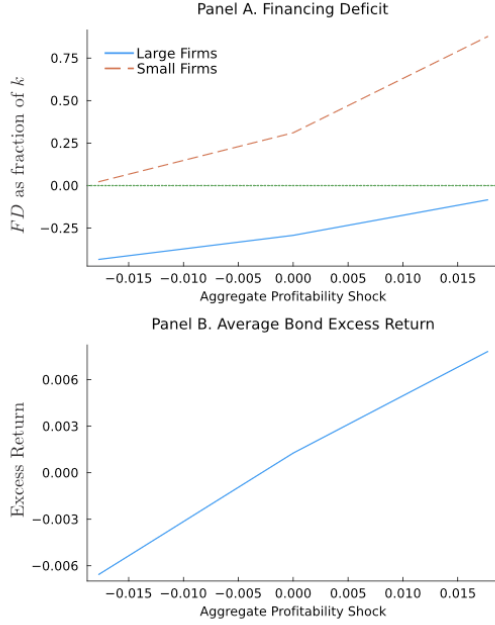
### Read:

- Table 1 shows the baseline fit is fairly good on the moments the authors care most about. For example, average risky assets are **0.0423 in the data** versus **0.0418 in the model**, average safe assets are **0.1432** versus **0.1411**, and average  $Q$  is **1.8837** versus **1.8802**.
- The estimated external financing wedge is economically meaningful:  $\hat{\xi} = 0.0951$ , so raising external funds costs about **9.5 cents per dollar**. The risky-bond trading cost is **5.77 bps**.
- Figure 3 shows how firms use their portfolios dynamically. In good aggregate states they liquidate both safe and risky assets to fund capital accumulation; in bad states they cut investment and are reluctant to trade the risky bond because of trading costs.
- Figure 4 ties the theory to cross-sectional heterogeneity. Small firms have financing deficits that rise more steeply in good aggregate states, while bond returns also rise in those states. This makes risky bonds especially valuable as hedges for small firms.

## 5.3 Table 2: simulated-data regression evidence

### Read:

- Table 2 is the “model-to-regression” bridge. It shows that if the model is simulated and then the authors run the same kind of second-stage regression they later estimate in the data, the financing deficit beta comes out strongly positive.
- In the full simulated sample, a one-standard-deviation increase in the financing deficit beta implies about **0.45% more risky assets relative to total assets** and **4.81% more risky assets in the financial portfolio**.



**Fig. 4.** Financing deficit and the incentives to invest in the risky security. Panel A plots the financing deficits, defined in Eq. (20) as  $FD = [i' + Adj^k - (1 - \tau)\pi - \tau\delta k]/k$ , of a small and a large firm, as functions of the aggregate profitability shock  $x$ . The small (large) firm is defined by setting capital  $k$  to a gridpoint below (above) the gridpoint closest to the simulated average  $k$ : specifically, the third quartile gridpoint below for small firms and the first quartile gridpoint above for large firms, based on the estimated parameters in Table 1. The financing deficit is positively correlated with  $x$  for both firms, with a stronger correlation for the small (dashed line) than the large (solid line) firm. Panel B plots the excess return of the risky bond against the aggregate profitability shock. The small firm (dashed line in Panel A) has stronger incentives to invest in the risky financial asset, as it provides higher returns in those states when the firm faces large financing deficits.

**Table 2**

Regressions of risky financial asset holdings using simulated data. This table reports the results of the regression in Eq. (23),  $FinAssets_{it} = \zeta_i + \phi\beta_{i,t-1}^{FD} + Controls_{i,t-1} + \varepsilon_{it}$ , using the simulated sample from the model. The dependent variable is the value of risky financial assets scaled either by total assets (variable *Risky* in odd-numbered columns) or total financial assets (variable *FinComp* in even-numbered columns). The independent variables,  $\beta_{i,t-1}^{FD}$  and  $FDResiVol_{i,t-1}$ , are the financing deficit beta and residual volatility computed using Eq. (22) and Eq. (24), respectively. The other control variables are *LQ*, *LProf*, *LSize*, and year fixed effects. Columns 1 and 2 report the results for the full simulated sample, Columns 3 and 4 for firms below yearly median firm size, and Columns 5 and 6 for firms above yearly median size. Significance levels at 10%, 5%, and 1% are denoted by \*, \*\*, and \*\*\*, respectively. We obtain the simulated sample by solving the model at the estimated parameters in Table 1 and simulating 50 samples of 3000 firms over 50 years. Variable definitions are provided in Table A.4. Appendix A presents the details of the numerical procedure used to solve the model.

| VARIABLES     | Full Sample           |                       | < Median Size         |                      | > Median Size         |                       |
|---------------|-----------------------|-----------------------|-----------------------|----------------------|-----------------------|-----------------------|
|               | (1)<br>Risky          | (2)<br>FinComp        | (3)<br>Risky          | (4)<br>FinComp       | (5)<br>Risky          | (6)<br>FinComp        |
| Beta          | 0.634***<br>(0.007)   | 5.748***<br>(0.049)   | 1.480***<br>(0.013)   | 11.639***<br>(0.065) | 0.332***<br>(0.008)   | 7.126***<br>(0.080)   |
| FDResiVol     | -0.446***<br>(0.002)  | -5.306***<br>(0.018)  | -0.607***<br>(0.004)  | -8.972***<br>(0.023) | 0.056***<br>(0.003)   | -2.032***<br>(0.028)  |
| LQ            | 3.306***<br>(0.010)   | 4.827***<br>(0.036)   | 3.505***<br>(0.014)   | 7.696***<br>(0.051)  | 2.672***<br>(0.027)   | -13.583***<br>(0.178) |
| LProf         | -20.854***<br>(0.038) | -22.451***<br>(0.231) | -32.653***<br>(0.082) | -7.065***<br>(0.389) | -15.382***<br>(0.077) | 3.156***<br>(0.437)   |
| LSize         | -0.190***<br>(0.007)  | -7.552***<br>(0.051)  | 1.703***<br>(0.023)   | -7.752***<br>(0.111) | 0.326***<br>(0.008)   | -8.833***<br>(0.083)  |
| Observations  | 7,500,000             | 7,500,000             | 3,750,000             | 3,750,000            | 3,750,000             | 3,750,000             |
| R-squared     | 0.364                 | 0.111                 | 0.370                 | 0.203                | 0.226                 | 0.082                 |
| Cluster Level | Firm                  | Firm                  | Firm                  | Firm                 | Firm                  | Firm                  |
| Year FEs      | Y                     | Y                     | Y                     | Y                    | Y                     | Y                     |

- The effect is larger for small firms, consistent with the theoretical prediction that precautionary motives are strongest when financing deficits are more exposed to aggregate conditions.

#### 5.4 Table 3 and Figures 5–6: sample facts and portfolio composition

**Table 3**

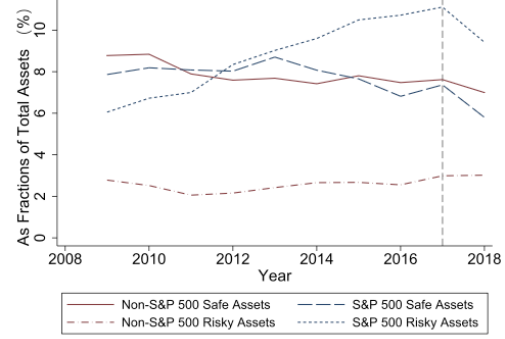
Summary statistics. This table reports the summary statistics of the variables used in the empirical analysis of Sections 4 and 5. Variables in Panel A are at the firm-year level, and in Panel B at the firm level. The measures of financial asset holdings used as dependent variables in the main regressions (Eq. (26)) are the fair value of risky and safe financial assets over total assets (*Risky* and *Safe*, respectively), and the fair value of risky financial assets over total financial assets (*FinComp*). The main independent variable of interest is the financing deficit beta (*Beta*), obtained by estimating for each firm the regression in Eq. (25),  $FD_{it} = \alpha_j + \beta_j^D R_{t-1}^D + \epsilon_{ijt}$ , where  $FD_{it}$  is the financing deficit of firm  $j$  in quarter  $t$ , and  $R_{t-1}^D$  is either the return on the Bloomberg US Aggregate Bond Market Index (for *BetaAgg*) or the reverse change in the 10-year Treasury bond yield (for *BetaTre*) during quarter  $t - 1$ . The financing deficit residual volatilities (*FDResiVolAgg* and *FDResiVolTre*) are the standard deviations of the residuals from the corresponding regressions. Other control variables are lagged Tobin's  $Q$  (*LQ*), profitability (*LProf*), and firm size (*LSize*). *InvTot* is the sum of tangible and intangible investment and *ExtFin* measures external financing—these variables are employed for the analysis in Section 5. The sample covers 2886 firms from 2009 to 2018. All variables in ratios are winsorized at the 1% and 99% percentiles. *BetaAgg* and *BetaTre* are winsorized at the 5% and 95% percentiles, and *FDResiVolAgg* and *FDResiVolTre* at the 95% percentile. Variable definitions are in Table A.4.

| Panel A. Firm-Year Level Variables |       |       |           |        |       |
|------------------------------------|-------|-------|-----------|--------|-------|
|                                    | N     | Mean  | Std. Dev. | P10    | P90   |
| Risky                              | 19367 | 0.042 | 0.108     | 0.000  | 0.138 |
| Safe                               | 19256 | 0.143 | 0.166     | 0.010  | 0.335 |
| FinComp                            | 19199 | 0.151 | 0.258     | 0.000  | 0.600 |
| LQ                                 | 19367 | 1.884 | 1.309     | 0.876  | 3.387 |
| LProf                              | 19367 | 0.076 | 0.181     | -0.098 | 0.226 |
| LSize                              | 19367 | 6.680 | 2.106     | 3.873  | 9.422 |
| InvTot                             | 19352 | 0.323 | 0.615     | 0.026  | 0.763 |
| ExtFin                             | 19367 | 0.072 | 0.200     | 0.000  | 0.203 |

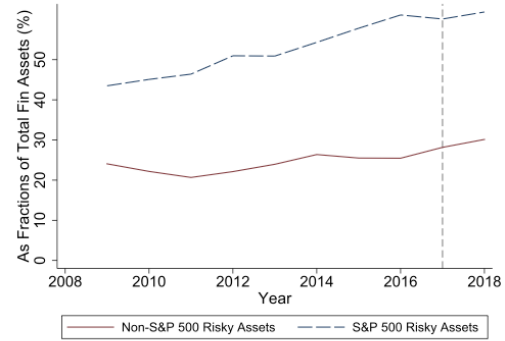
  

| Panel B. Firm Level Variables |      |        |           |         |        |
|-------------------------------|------|--------|-----------|---------|--------|
|                               | N    | Mean   | Std. Dev. | P10     | P90    |
| BetaAgg                       | 2886 | 0.409  | 3.208     | -3.741  | 5.178  |
| BetaTre                       | 2886 | 1.010  | 12.255    | -15.293 | 20.950 |
| FDResiVolAgg                  | 2886 | 27.580 | 29.428    | 3.872   | 83.861 |
| FDResiVolTre                  | 2886 | 27.744 | 29.797    | 3.850   | 82.872 |

Panel A. Risky and Safe Financial Assets of S&P 500 and Non-S&P 500 Firms



Panel B. Fractions of Risky Financial Assets of S&P 500 and Non-S&P 500 Firms



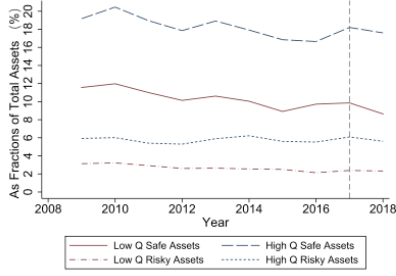
**Fig. 5.** Savings portfolio composition of S&P 500 and non-S&P 500 firms. This figure plots the aggregate value of safe and risky financial assets scaled by aggregate total assets (Panel A), and the fraction of aggregate risky over total financial assets (Panel B) for both S&P 500 and non-S&P 500 firms. The sample covers the years 2009 to 2018.

#### Read:

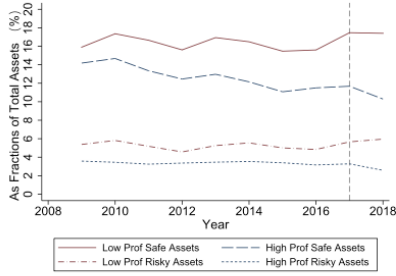
- Table 3 shows that risky financial assets are not a tiny residual category. On average they are **4.2% of total assets**, while safe assets are **14.3% of total assets**.
- Figure 5 shows that S&P 500 firms hold very large risky financial portfolios, especially before the 2017 tax reform. Non-S&P 500 firms hold less risky financial wealth, but the amounts are still economically meaningful.
- Figure 6 shows where these holdings are concentrated: firms with high  $Q$ , low profitability, and high-tech orientation hold more financial assets. That pattern fits the paper's story that growth opportunities and financing frictions both strengthen precautionary motives.



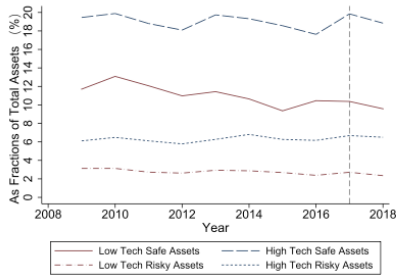
Panel A. Risky and Safe Financial Assets: High- vs Low-Q Firms



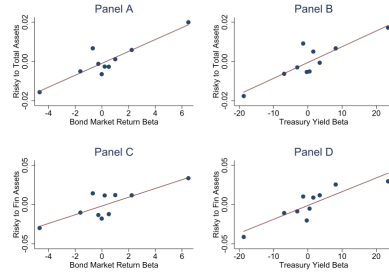
Panel B. Risky and Safe Financial Assets: High- vs Low-Prof Firms



Panel C. Risky and Safe Financial Assets: High- vs Low-Tech Firms



**Fig. 6.** Savings portfolio composition by firm characteristics and industry. This figure shows average safe and risky financial assets as a percentage of total assets, split by firm characteristics: Tobin's Q (Panel A), profitability (Panel B), and industry classification (Panel C). High (Low) Q firms are those above (below) the median Q within each year; high (low) profitability firms are defined similarly. High-tech industries are identified using the classification in [Francis and Schipper \(1999\)](#) based on 2-digit SIC codes. The sample covers the years 2009 to 2018.



**Fig. 7.** Financing deficit beta and corporate financial assets. This figure plots the relationship between the financing deficit beta and risky financial asset holdings. The figure is constructed using a two-step sorting procedure. For each firm, we estimate the time-series regression in Eq. (25),  $FD_{it} = \alpha_i + \beta_i^B R_{Bt}^e + \epsilon_{it}$ , where  $FD_{it}$  is the financing deficit of firm  $i$  in quarter  $t$ , and  $R_{Bt}^e$  is either the return on the Bloomberg US Aggregate Bond Market Index (for  $BetaAgg$ ) or the reverse change in the 10-year Treasury bond yield (for  $BetaTre$ ) in quarter  $t - 1$ . The financing deficit residual volatilities ( $FDResVolAgg$  and  $FDResVolTre$ ) are the standard deviations of the residuals from the corresponding regressions. In the first step, each year we sort firms into deciles based on  $FDResVolAgg$  (Panels A and C) or  $FDResVolTre$  (Panels B and D), and we compute, by subtracting the year-decile mean, the residuals in firms' risky financial asset holdings. This step teases out the effects of volatility on saving behavior. The second step is to sort firms each year into deciles based on  $BetaAgg$  (Panels A and C) or  $BetaTre$  (Panels B and D), and compute the decile means of the residuals from the first step. The graphs plot the decile means from the second step. Panels A and B plot decile means of the residuals in risky financial assets to total assets against the financing deficit beta, while Panels C and D use risky financial assets scaled by total financial assets. The sample period is 2009 to 2018.

## 5.5 Figure 7 and Table 4: the main empirical evidence

Read:

- Figure 7 shows a monotone positive relation between the financing deficit beta and risky financial asset holdings after first purging the effect of residual financing-deficit volatility. This is the cleanest visual version of the main empirical result.
- Table 4 confirms the pattern in regressions. The coefficients on the financing deficit beta are positive and highly significant when the dependent variable is either risky assets over total assets or risky over total financial assets.
- Using the Agg-based beta, the coefficient is **0.252** for Risky and **0.542** for FinComp. Using the Treasury-based beta, the corresponding coefficients are **0.072** and **0.165**.
- In economic terms, a one-standard-deviation increase in the beta implies about **0.81% to 0.88% more risky financial assets relative to total assets**. Since the sample mean of Risky is only 4.2%, this is a big effect.
- The coefficients for **Safe** are weak and mostly insignificant. That matters: it says the beta is not just picking up a generic desire to save more; it is specifically related to the desire to hold *risky* financial assets.

**Table 4**  
Regressions of corporate financial asset holdings—main results. This table reports the results of the regression in Eq. (26) estimated for the full sample,  $FinAssets_{jt} = \zeta_{ic,t} + \phi \beta_{jt}^D + Controls_{jt-1} + \epsilon_{jt}$ , where  $j$  and  $t$  index firm and year, respectively;  $FinAssets_{jt}$  is either the fair value of risky financial assets over total assets (*Risky* in Columns 1 and 2), safe financial assets over total assets (*Safe* in Columns 3 and 4), or the fair value of risky financial assets over total financial assets (*FinComp* in Columns 5 and 6);  $\zeta_{ic,t}$  are 2-digit SIC code industry  $\times$  year fixed effects; the control variables are  $LQ$ ,  $LProf$ ,  $LSize$ , and  $FDResiVol$ ; and  $\beta_{jt}^D$ , the financing deficit beta, is estimated for each firm from the time-series regression in Eq. (25),  $FD_{jt} = \alpha_j + \beta_{jt}^D R_{t-1}^H + \epsilon_{jt}$ , where  $FD_{jt}$  is the financing deficit of firm  $j$  in quarter  $t$ ;  $R_{t-1}^H$  is either the return on the Bloomberg US Aggregate Bond Market Index ("Agg Bond") or the reverse change in the 10-year Treasury bond yield ("Treasury") during quarter  $t-1$ ; and  $FDResiVol$  is the standard deviation of the financing deficit residuals,  $\epsilon_{jt}$ , from the corresponding regressions. Standard errors in parentheses are clustered at the industry level and adjusted using the procedure in [Murphy and Topel \(1985\)](#) to account for sampling error in the first stage. Significance levels at 10%, 5%, and 1% are denoted by \*, \*\*, and \*\*\*, respectively. Variable definitions are in [Table A.4](#).

| VARIABLES             | (1)<br>Risky        | (2)<br>Risky        | (3)<br>Safe          | (4)<br>Safe          | (5)<br>FinComp      | (6)<br>FinComp      |
|-----------------------|---------------------|---------------------|----------------------|----------------------|---------------------|---------------------|
| Beta                  | 0.252***<br>(0.040) | 0.072***<br>(0.013) | 0.071<br>(0.091)     | 0.039*<br>(0.021)    | 0.542***<br>(0.126) | 0.165***<br>(0.035) |
| FDResiVol             | 0.034**<br>(0.016)  | 0.035**<br>(0.016)  | 0.042*<br>(0.021)    | 0.041*<br>(0.021)    | 0.037<br>(0.027)    | 0.039<br>(0.027)    |
| LQ                    | 1.534***<br>(0.267) | 1.544***<br>(0.268) | 3.869***<br>(0.304)  | 3.866***<br>(0.304)  | 1.392***<br>(0.422) | 1.411***<br>(0.415) |
| LProf                 | -2.828<br>(1.795)   | -2.828<br>(1.778)   | -8.814**<br>(3.376)  | -8.766**<br>(3.355)  | -3.982<br>(3.009)   | -3.973<br>(2.959)   |
| LSize                 | 0.297**<br>(0.132)  | 0.295**<br>(0.131)  | -1.146***<br>(0.108) | -1.147***<br>(0.107) | 1.288***<br>(0.217) | 1.282***<br>(0.217) |
| Observations          | 19,367              | 19,367              | 19,256               | 19,256               | 19,199              | 19,199              |
| R-squared             | 0.106               | 0.107               | 0.276                | 0.276                | 0.062               | 0.063               |
| Beta Type             | Agg Bond            | Treasury            | Agg Bond             | Treasury             | Agg Bond            | Treasury            |
| Cluster Level         | Industry            | Industry            | Industry             | Industry             | Industry            | Industry            |
| Ind $\times$ Year FEs | Y                   | Y                   | Y                    | Y                    | Y                   | Y                   |

**Table 5**

Interest-rate risk and risky financial asset holdings. This table reports the results of the regression in Eq. (26) estimated for the full sample,  $FinAssets_{jt} = \zeta_{ic,t} + \phi \beta_{jt}^D + Controls_{jt-1} + \epsilon_{jt}$ , where  $j$  and  $t$  index firm and year, respectively;  $FinAssets_{jt}$  is either the fair value of risky financial assets over total assets (*Risky* in Panel A), or the fair value of risky financial assets over total financial assets (*FinComp* in Panel B);  $\zeta_{ic,t}$  are 2-digit SIC code industry  $\times$  year fixed effects; the control variables are  $LQ$ ,  $LProf$ ,  $LSize$ , and  $FDResiVol$ ; and  $\beta_{jt}^D$ , the financing deficit beta, is estimated for each firm from the time-series regression in Eq. (25),  $FD_{jt} = \alpha_j + \beta_{jt}^D R_{t-1}^H + \epsilon_{jt}$ , where  $FD_{jt}$  is the financing deficit of firm  $j$  in quarter  $t$ ; and  $R_{t-1}^H$  is the reverse change in the Fed Funds rate (Column 1), and the 1-year (Column 2), 2-year (Column 3), 3-year (Column 4), 5-year (Column 5) and 10-year (Column 6) Treasury bond yield in quarter  $t-1$ , respectively. Standard errors in parentheses are clustered at the industry level and adjusted using the procedure in [Murphy and Topel \(1985\)](#) to account for sampling error in the first stage. Significance levels at 10%, 5%, and 1% are denoted by \*, \*\*, and \*\*\*, respectively. Variable definitions are in [Table A.4](#).

| Panel A. Risky Financial Assets over Total Assets |                     |                     |                     |                     |                     |                     |
|---------------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| VARIABLES                                         | (1)<br>Risky        | (2)<br>Risky        | (3)<br>Risky        | (4)<br>Risky        | (5)<br>Risky        | (6)<br>Risky        |
| Beta                                              | 0.023***<br>(0.006) | 0.043***<br>(0.007) | 0.049***<br>(0.008) | 0.057***<br>(0.008) | 0.078***<br>(0.010) | 0.072***<br>(0.013) |
| Observations                                      | 19,367              | 19,367              | 19,367              | 19,367              | 19,367              | 19,367              |
| R-squared                                         | 0.105               | 0.111               | 0.109               | 0.109               | 0.109               | 0.107               |
| Beta Type                                         | Fed Funds           | 1-Y Tre             | 2-Y Tre             | 3-Y Tre             | 5-Y Tre             | 10-Y Tre            |
| Cluster Level                                     | Industry            | Industry            | Industry            | Industry            | Industry            | Industry            |
| Controls                                          | Y                   | Y                   | Y                   | Y                   | Y                   | Y                   |
| Ind $\times$ Year FEs                             | Y                   | Y                   | Y                   | Y                   | Y                   | Y                   |

| Panel B. Risky Financial Assets over Total Financial Assets |                  |                     |                     |                     |                     |                     |
|-------------------------------------------------------------|------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| VARIABLES                                                   | (1)<br>FinComp   | (2)<br>FinComp      | (3)<br>FinComp      | (4)<br>FinComp      | (5)<br>FinComp      | (6)<br>FinComp      |
| Beta                                                        | 0.019<br>(0.013) | 0.056***<br>(0.015) | 0.074***<br>(0.018) | 0.090***<br>(0.020) | 0.145***<br>(0.029) | 0.165***<br>(0.035) |
| Observations                                                | 19,199           | 19,199              | 19,199              | 19,199              | 19,199              | 19,199              |
| R-squared                                                   | 0.059            | 0.061               | 0.061               | 0.061               | 0.063               | 0.063               |
| Beta Type                                                   | Fed Funds        | 1-Y Tre             | 2-Y Tre             | 3-Y Tre             | 5-Y Tre             | 10-Y Tre            |
| Cluster Level                                               | Industry         | Industry            | Industry            | Industry            | Industry            | Industry            |
| Controls                                                    | Y                | Y                   | Y                   | Y                   | Y                   | Y                   |
| Ind $\times$ Year FEs                                       | Y                | Y                   | Y                   | Y                   | Y                   | Y                   |

## 5.6 Tables 5 and 6: which risks matter?

Read:

- Table 5 asks which part of the term structure matters. The coefficient on the beta rises with the maturity of the reference rate: for **Risky**, it increases from **0.023** with the Fed Funds rate to **0.078** with the 5-year Treasury and **0.072** with the 10-year Treasury.
- For the portfolio-composition measure **FinComp**, the maturity pattern is even stronger: the Fed Funds beta is insignificant, while the 10-year Treasury beta is **0.165**. So the paper's risky-assets story is mainly a **medium- to long-term interest-rate-risk** story.
- Table 6 considers alternative risk factors. Equity-market beta is negative when entered alone, which is plausible because the firm's risky financial assets are mostly bond-like rather than equity-like. But once the Treasury beta is added, the equity effect becomes much weaker.
- Betas based on liquidity and high-yield spreads are not robust. The investment-grade spread has some explanatory power, but the Treasury-based interest-rate beta remains the most stable predictor throughout.

**Table 6**  
Alternative risk factors and risky financial asset holdings. This table reports the results of the regression in Eq. (26) estimated for the full sample,  $FinAssets_{jt} = \zeta_{m,t} + \phi\beta^R + Controls_{j,t-1} + \varepsilon_{jt}$ , where  $j$  and  $t$  index firm and year, respectively;  $FinAssets_{jt}$  is either the fair value of risky financial assets over total assets (*Risky* in Panel A), or the fair value of risky financial assets over total financial assets (*FinComp* in Panel B);  $\zeta_{m,t}$  are 2-digit SIC code industry  $\times$  year fixed effects; the control variables are  $LO$ ,  $LP$ ,  $LF$ ,  $LS$ ,  $LI$ , and  $FD$ ;  $\beta^R$  is the alternative financing deficit beta measure, is estimated for each firm from the time-series regression in Eq. (25),  $FD_{jt} = \alpha + \beta^R R^e_{jt} + \varepsilon_{jt}$ , where  $FD_{jt}$  is the financing deficit of firm  $j$  in quarter  $t$ ; and  $R^e_{jt}$  is either the return on the S&P 500 index (Columns 1 and 2), the liquidity factor return from Pastor and Stambaugh (2003) (Columns 3 and 4), the reverse change in the investment-grade corporate bond spread over the Fed Funds rate (Columns 5 and 6), or in the high-yield corporate bond spread (Columns 7 and 8), in quarter  $t-1$ . Even-numbered columns include as a control the financing deficit beta estimated using the reverse change in the 10-year Treasury bond yield,  $BetaTre$ . Standard errors in parentheses are clustered at the industry level and adjusted using the procedure in Murphy and Topel (1985) to account for sampling error in the first stage. Significance levels at 10%, 5%, and 1% are denoted by \*, \*\*, and \*\*\*, respectively. Variable definitions are in Table A.4.

| Panel A. Risky Financial Assets over Total Assets |                      |                     |                   |                     |                  |                     |                    |                     |
|---------------------------------------------------|----------------------|---------------------|-------------------|---------------------|------------------|---------------------|--------------------|---------------------|
| VARIABLES                                         | (1)<br>Risky         | (2)<br>Risky        | (3)<br>Risky      | (4)<br>Risky        | (5)<br>Risky     | (6)<br>Risky        | (7)<br>Risky       | (8)<br>Risky        |
| BetaAlt                                           | -1.025***<br>(0.260) | -0.656**<br>(0.252) | -0.424<br>(0.421) | -0.305<br>(0.448)   | 0.010<br>(0.021) | 0.046**<br>(0.019)  | -0.077*<br>(0.045) | 0.018<br>(0.044)    |
| BetaTre                                           |                      | 0.057***<br>(0.012) |                   | 0.069***<br>(0.014) |                  | 0.084***<br>(0.013) |                    | 0.074***<br>(0.013) |
| Observations                                      | 19,367               | 19,367              | 19,367            | 19,367              | 19,367           | 19,367              | 19,367             | 19,367              |
| R-squared                                         | 0.106                | 0.109               | 0.103             | 0.108               | 0.103            | 0.109               | 0.103              | 0.108               |
| BetaAlt Type                                      | Equity               | Equity              | Liquidity         | Liquidity           | Inv Spread       | Inv Spread          | HY Spread          | HY Spread           |
| Cluster Level                                     | Industry             | Industry            | Industry          | Industry            | Industry         | Industry            | Industry           | Industry            |
| Controls                                          | Y                    | Y                   | Y                 | Y                   | Y                | Y                   | Y                  | Y                   |
| Ind $\times$ Year FEs                             | Y                    | Y                   | Y                 | Y                   | Y                | Y                   | Y                  | Y                   |

| Panel B. Risky Financial Assets over Total Financial Assets |                      |                     |                   |                     |                   |                     |                      |                     |
|-------------------------------------------------------------|----------------------|---------------------|-------------------|---------------------|-------------------|---------------------|----------------------|---------------------|
| VARIABLES                                                   | (1)<br>FinComp       | (2)<br>FinComp      | (3)<br>FinComp    | (4)<br>FinComp      | (5)<br>FinComp    | (6)<br>FinComp      | (7)<br>FinComp       | (8)<br>FinComp      |
| BetaAlt                                                     | -2.499***<br>(0.690) | -1.676**<br>(0.641) | -0.203<br>(0.799) | 0.085<br>(0.855)    | -0.036<br>(0.048) | 0.041<br>(0.046)    | -0.279***<br>(0.106) | -0.080<br>(0.100)   |
| BetaTre                                                     |                      | 0.126***<br>(0.030) |                   | 0.165***<br>(0.036) |                   | 0.175***<br>(0.035) |                      | 0.154***<br>(0.034) |
| Observations                                                | 19,199               | 19,199              | 19,199            | 19,199              | 19,199            | 19,199              | 19,199               | 19,199              |
| R-squared                                                   | 0.062                | 0.064               | 0.059             | 0.063               | 0.059             | 0.063               | 0.060                | 0.063               |
| BetaAlt Type                                                | Equity               | Equity              | Liquidity         | Liquidity           | Inv Spread        | Inv Spread          | HY Spread            | HY Spread           |
| Cluster Level                                               | Industry             | Industry            | Industry          | Industry            | Industry          | Industry            | Industry             | Industry            |
| Controls                                                    | Y                    | Y                   | Y                 | Y                   | Y                 | Y                   | Y                    | Y                   |
| Ind $\times$ Year FEs                                       | Y                    | Y                   | Y                 | Y                   | Y                 | Y                   | Y                    | Y                   |

**Table 7**  
Structural estimation results—small vs large firms. Panel A reports the sample and simulated moments. Columns 1–3 report, for small firms, the empirical moments targeted in the structural estimation, the corresponding simulated moments generated from 50 simulations of 3000 firms over 50 years, and the  $t$ -statistics for the differences between empirical and simulated moments, respectively. Columns 4–6 report the same set of statistics for large firms. Panels B and C report the estimated structural parameters for small and large firms, respectively. The parameters  $\rho_i$  and  $\sigma_i$  are the persistence and conditional standard deviation of the idiosyncratic profitability shock, respectively, and they are obtained from an autocorrelation regression of TFP using data from Imrohorglu and Tuzel (2014);  $\alpha$  is the curvature of the profit function;  $\psi_i$  is the capital adjustment cost parameter;  $\xi$  is the external financing cost parameter; and  $w_i$  denotes the bond trading cost. Small and large firms are defined as firms below and above the yearly median of firm size distribution, respectively. Standard errors are clustered at the firm level and adjusted for sampling errors in the first stage. Section 4.1 describes the sample construction. Moment definitions are provided in Table A.3. Appendix A presents details of the numerical procedure used to solve the model and of the estimation approach.

| Panel A. Moments                 |             |              |                   |             |              |                   |
|----------------------------------|-------------|--------------|-------------------|-------------|--------------|-------------------|
|                                  | Small Firms |              |                   | Large Firms |              |                   |
|                                  | (1)<br>Data | (2)<br>Model | (3)<br>$t$ -stats | (4)<br>Data | (5)<br>Model | (6)<br>$t$ -stats |
| Average Risky Financial Assets   | 0.0505      | 0.0508       | -0.61             | 0.0341      | 0.0390       | -16.04            |
| Average Safe Financial Assets    | 0.1840      | 0.1842       | -0.81             | 0.1020      | 0.1087       | -15.14            |
| Average Investment               | 0.1328      | 0.1382       | -5.18             | 0.1131      | 0.1318       | -8.75             |
| Average Q                        | 2.0205      | 2.0209       | -0.05             | 1.7468      | 1.9633       | -24.55            |
| Average Profitability            | 0.0626      | 0.1687       | -5.20             | 0.3763      | 0.1371       | 6.62              |
| Median Distribution              | 0.0004      | 0.0035       | -0.15             | 0.0038      | 0.0024       | 4.19              |
| Variance of Profitability        | 0.0119      | 0.0074       | 1.97              | 0.0038      | 0.0055       | -0.82             |
| Variance of Distribution         | 0.0379      | 0.0261       | 2.82              | 0.0140      | 0.0372       | -13.83            |
| Autocorrelation of Profitability | 0.7454      | 0.6399       | 1.48              | 0.9242      | 0.7258       | 1.13              |
| Autocorrelation of Investment    | 0.6388      | 0.4865       | 2.36              | 0.7702      | 0.5683       | 1.54              |
| Autocorrelation of Distribution  | 0.6047      | 0.4308       | 2.95              | 0.3860      | 0.5221       | -0.64             |

| Panel B. Small Firms |          |            |          |          |        |             |
|----------------------|----------|------------|----------|----------|--------|-------------|
| Parameter            | $\rho_i$ | $\sigma_i$ | $\alpha$ | $\psi_i$ | $\xi$  | $w_i$ (bps) |
| Value                | 0.7410   | 0.3783     | 0.8141   | 0.2677   | 0.1016 | 6.6538      |
| Standard Error       | 0.0161   | 0.0116     | 0.0185   | 0.0395   | 0.0061 | 0.6099      |

| Panel C. Large Firms |          |            |          |          |        |             |
|----------------------|----------|------------|----------|----------|--------|-------------|
| Parameter            | $\rho_i$ | $\sigma_i$ | $\alpha$ | $\psi_i$ | $\xi$  | $w_i$ (bps) |
| Value                | 0.7944   | 0.3026     | 0.8801   | 0.4851   | 0.0693 | 3.4985      |
| Standard Error       | 0.0212   | 0.0110     | 0.2325   | 0.4114   | 0.0054 | 8.6933      |

## 5.7 Tables 7–11: heterogeneity across firm types

Read:

- The structural subsample estimates in Tables 7–9 line up with the economic story. Small, high-volatility, and R&D-intensive firms have higher estimated financing frictions and therefore stronger precautionary saving motives.
- For example, the estimated external financing cost parameter is **0.1016** for small firms versus **0.0693** for large firms; **0.1318** for high-volatility firms versus **0.0965** for low-volatility firms; and **0.1397** for high-R&D firms versus **0.0679** for low-R&D firms.
- Table 10 shows that the simulated-data regressions reproduce this heterogeneity: the beta coefficients are larger in the subsamples that the model identifies as more financially constrained.

- Table 11 shows the same qualitative pattern in the real data. For **small firms**, a one-standard-deviation increase in the financing deficit beta raises risky assets by about **1.00%** of total assets using the Agg beta and **0.86%** using the Treasury beta. For risky assets as a share of total financial assets, the corresponding increases are about **1.75%** and **1.77%**.
- For **high-volatility firms**, the estimated effects are again strong: about **0.85%** and **0.83%** more risky assets relative to total assets, and about **1.82%** and **1.95%** more risky assets as a fraction of total financial assets.
- For **R&D splits**, the coefficients remain positive in both groups, but the pattern is strongest for the high-R&D firms. So precautionary motives do not disappear elsewhere, but they are clearest where external financing is especially costly.

Table 8

Structural estimation results—high vs low volatility firms. Panel A reports the sample and simulated moments. Columns 1-3 report, for high volatility firms, the empirical moments targeted in the structural estimation, the corresponding simulated moments generated from 50 simulations of 3000 firms over 50 years, and the  $t$ -statistics for the differences between empirical and simulated moments, respectively. Columns 4-6 report the same set of statistics for low volatility firms. Panels B and C report the estimated structural parameters for high and low volatility firms, respectively. The parameters  $\rho_e$  and  $\sigma_e$  are the persistence and conditional standard deviation of the idiosyncratic profitability shock, respectively, and they are obtained from an autocorrelation regression of TFP using data from Imrohoroglu and Tuzel (2014);  $\alpha$  is the curvature of the profit function;  $\psi_e$  is the capital adjustment cost parameter;  $\zeta$  is the external financing cost parameter; and  $\psi_f$  denotes the bond trading cost. High and low volatility firms are defined as firms above and below the median of firm-level financing deficit volatility, respectively. Standard errors are clustered at the firm level and adjusted for sampling errors in the first stage. Section 4.1 describes the sample construction. Moment definitions are provided in Table A.3. Appendix A presents details of the numerical procedure used to solve the model and of the estimation approach.

| Panel A. Moments                 |                       |              |                |                      |              |                |
|----------------------------------|-----------------------|--------------|----------------|----------------------|--------------|----------------|
|                                  | High Volatility Firms |              |                | Low Volatility Firms |              |                |
|                                  | (1)<br>Data           | (2)<br>Model | (3)<br>t-stats | (4)<br>Data          | (5)<br>Model | (6)<br>t-stats |
| Average Risky Financial Assets   | 0.0554                | 0.0576       | -43.78         | 0.0293               | 0.0292       | 0.57           |
| Average Safe Financial Assets    | 0.1755                | 0.1837       | -28.96         | 0.1111               | 0.1128       | -15.33         |
| Average Investment               | 0.1479                | 0.1278       | 8.19           | 0.0980               | 0.1180       | -11.79         |
| Average Q                        | 2.0998                | 2.2044       | -17.21         | 1.6678               | 1.7520       | -9.99          |
| Average Profitability            | 0.2348                | 0.1694       | 2.79           | 0.2044               | 0.1629       | 5.17           |
| Median Distribution              | 0.0025                | 0.0035       | -0.05          | 0.0250               | 0.0042       | 4.73           |
| Variance of Profitability        | 0.0105                | 0.0090       | 0.57           | 0.0099               | 0.0058       | 0.07           |
| Variance of Distribution         | 0.0423                | 0.0191       | 9.93           | 0.0120               | 0.0198       | -8.61          |
| Autocorrelation of Profitability | 0.7731                | 0.7054       | 0.93           | 0.6909               | 0.7514       | -0.41          |
| Autocorrelation of Investment    | 0.6791                | 0.5548       | 1.82           | 0.6086               | 0.5723       | 0.46           |
| Autocorrelation of Distribution  | 0.5621                | 0.4438       | 2.46           | 0.3182               | 0.4832       | -1.17          |

| Panel B. High Volatility Firms |          |            |          |          |         |                |
|--------------------------------|----------|------------|----------|----------|---------|----------------|
| Parameter                      | $\rho_e$ | $\sigma_e$ | $\alpha$ | $\psi_e$ | $\zeta$ | $\psi_f$ (bps) |
| Value                          | 0.8058   | 0.3762     | 0.8079   | 0.4726   | 0.1318  | 6.9397         |
| Standard Error                 | 0.0166   | 0.0148     | 0.0317   | 0.1216   | 0.0141  | 1.2929         |

| Panel C. Low Volatility Firms |          |            |          |          |         |                |
|-------------------------------|----------|------------|----------|----------|---------|----------------|
| Parameter                     | $\rho_e$ | $\sigma_e$ | $\alpha$ | $\psi_e$ | $\zeta$ | $\psi_f$ (bps) |
| Value                         | 0.8013   | 0.3006     | 0.8974   | 0.8822   | 0.0965  | 7.0583         |
| Standard Error                | 0.0200   | 0.0086     | 0.0218   | 0.0576   | 0.0031  | 4.0444         |

Table 9

Structural estimation results—high vs low R&D firms. Panel A reports the sample and simulated moments. Columns 1-3 report, for high R&D firms, the empirical moments targeted in the structural estimation, the corresponding simulated moments generated from 50 simulations of 3000 firms over 50 years, and the  $t$ -statistics for the differences between empirical and simulated moments, respectively. Columns 4-6 report the same set of statistics for low R&D firms. Panels B and C report the estimated structural parameters for high and low R&D firms, respectively. The parameters  $\rho_e$  and  $\sigma_e$  are the persistence and conditional standard deviation of the idiosyncratic profitability shock, respectively, and they are obtained from an autocorrelation regression of TFP using data from Imrohoroglu and Tuzel (2014);  $\alpha$  is the curvature of the profit function;  $\psi_e$  is the capital adjustment cost parameter;  $\zeta$  is the external financing cost parameter; and  $\psi_f$  denotes the bond trading cost. High and low R&D firms are defined as firms with positive and zero R&D expenditures during the sample period, respectively. Standard errors are clustered at the firm level and adjusted for sampling errors in the first stage. Section 4.1 describes the sample construction. Moment definitions are provided in Table A.3. Appendix A presents details of the numerical procedure used to solve the model and of the estimation approach.

| Panel A. Moments                 |                |              |                |               |              |                |
|----------------------------------|----------------|--------------|----------------|---------------|--------------|----------------|
|                                  | High R&D Firms |              |                | Low R&D Firms |              |                |
|                                  | (1)<br>Data    | (2)<br>Model | (3)<br>t-stats | (4)<br>Data   | (5)<br>Model | (6)<br>t-stats |
| Average Risky Financial Assets   | 0.0599         | 0.0600       | -0.48          | 0.0246        | 0.0256       | -113.30        |
| Average Safe Financial Assets    | 0.1880         | 0.1888       | -2.81          | 0.0981        | 0.1292       | -102.82        |
| Average Investment               | 0.1253         | 0.1274       | -2.66          | 0.1205        | 0.1382       | -6.08          |
| Average Q                        | 2.1271         | 2.1670       | -19.71         | 1.6376        | 2.1225       | -140.93        |
| Average Profitability            | 0.2015         | 0.1706       | 1.41           | 0.2379        | 0.1579       | 5.63           |
| Median Distribution              | 0.0196         | 0.0039       | 0.74           | 0.0170        | 0.0026       | 1.57           |
| Variance of Profitability        | 0.0070         | 0.0082       | -0.47          | 0.0093        | 0.0059       | 2.19           |
| Variance of Distribution         | 0.0322         | 0.0183       | 3.52           | 0.0220        | 0.0437       | -6.25          |
| Autocorrelation of Profitability | 0.7978         | 0.7083       | 1.05           | 0.6986        | 0.7123       | -0.14          |
| Autocorrelation of Investment    | 0.6412         | 0.5543       | 1.02           | 0.6982        | 0.5662       | 1.95           |
| Autocorrelation of Distribution  | 0.5450         | 0.4280       | 1.36           | 0.5786        | 0.5077       | 0.74           |

| Panel B. High R&D Firms |          |            |          |          |         |                |
|-------------------------|----------|------------|----------|----------|---------|----------------|
| Parameter               | $\rho_e$ | $\sigma_e$ | $\alpha$ | $\psi_e$ | $\zeta$ | $\psi_f$ (bps) |
| Value                   | 0.8086   | 0.3552     | 0.8063   | 0.4429   | 0.1397  | 6.8493         |
| Standard Error          | 0.0154   | 0.0128     | 0.0284   | 0.1061   | 0.0099  | 1.1019         |

| Panel C. Low R&D Firms |          |            |          |          |         |                |
|------------------------|----------|------------|----------|----------|---------|----------------|
| Parameter              | $\rho_e$ | $\sigma_e$ | $\alpha$ | $\psi_e$ | $\zeta$ | $\psi_f$ (bps) |
| Value                  | 0.7971   | 0.3176     | 0.8585   | 0.3807   | 0.0679  | 4.7340         |
| Standard Error         | 0.0220   | 0.0111     | 0.0161   | 0.0507   | 0.0012  | 1.0004         |

## 5.8 Figure 8 and Table 12: the value of access to risky financial assets

Read:

- Figure 8 varies the trading cost  $\psi_s$ . As risky-asset trading becomes more expensive, firms substitute toward safe assets. Investment and firm value both fall, though the effects are not huge because safe and risky assets are partial substitutes.
- Table 12 is the main counterfactual. When the model forbids firms from holding risky financial assets, they increase safe savings somewhat but *total* savings still fall, investment falls, and reliance on external financing rises.
- Quantitatively, the no-risky-assets economy has lower capital, lower profits, lower financial-asset cash flows, and **about 0.35% lower firm value**. Issuance frequency rises by about **4.70%**.
- The losses are larger in **booms** than in busts, and larger for **small firms** than for large firms. This fits the model's mechanism perfectly: the risky asset is most valuable precisely when the firm wants to fund expansion.

**Table 10**

Regressions of risky financial asset holdings—simulated subsamples. This table reports the results of the regression in Eq. (23),  $FinAssets_{i,t} = \zeta_i + \phi \beta_{i,t}^{FD} + Control_{i,t} + \epsilon_{i,t}$ , using the simulated samples from the model. The dependent variable is the value of risky financial assets scaled either by total assets (variable *Risky* in Panel A) or total financial assets (variable *FinComp* in Panel B). The independent variable  $\beta_{i,t}^{FD}$  is the financing deficit beta computed using Eq. (22). Control variables include *LQ*, *LProf*, *LSize*, *FDResVol* computed using Eq. (24), and year fixed effects. In both panels, Columns 1 to 6 report the regression results using subsamples simulated from the structural estimates for small (Panel B in Table 7), large (Panel C in Table 7), high volatility (Panel B in Table 8), low volatility (Panel C in Table 8), high R&D (Panel B in Table 9), and low R&D (Panel C in Table 9) firms, respectively. Significance levels at 10%, 5%, and 1% are denoted by \*, \*\*, and \*\*\*, respectively. We obtain the simulated samples by solving the model at the estimated parameters and simulating 50 samples of 3000 firms over 50 years. Variable definitions are provided in Table A.4. Appendix A presents the details of the numerical procedure used to solve the model.

| Panel A. Risky Financial Assets over Total Assets |                       |                       |                          |                         |                          |                         |
|---------------------------------------------------|-----------------------|-----------------------|--------------------------|-------------------------|--------------------------|-------------------------|
| VARIABLES                                         | Small<br>(1)<br>Risky | Large<br>(2)<br>Risky | High-Vol<br>(3)<br>Risky | Low-Vol<br>(4)<br>Risky | High-R&D<br>(5)<br>Risky | Low-R&D<br>(6)<br>Risky |
| Beta                                              | 1.119***<br>(0.007)   | 0.381***<br>(0.007)   | 2.117***<br>(0.008)      | 0.267***<br>(0.006)     | 2.405***<br>(0.008)      | 0.924***<br>(0.006)     |
| Observations                                      | 7,500,000             | 7,500,000             | 7,500,000                | 7,500,000               | 7,500,000                | 7,500,000               |
| R-squared                                         | 0.389                 | 0.339                 | 0.388                    | 0.347                   | 0.402                    | 0.345                   |
| Cluster Level                                     | Firm                  | Firm                  | Firm                     | Firm                    | Firm                     | Firm                    |
| Controls                                          | Y                     | Y                     | Y                        | Y                       | Y                        | Y                       |
| Year FEs                                          | Y                     | Y                     | Y                        | Y                       | Y                        | Y                       |

| Panel B. Risky Financial Assets over Total Financial Assets |                         |                         |                            |                           |                            |                           |
|-------------------------------------------------------------|-------------------------|-------------------------|----------------------------|---------------------------|----------------------------|---------------------------|
| VARIABLES                                                   | Small<br>(1)<br>FinComp | Large<br>(2)<br>FinComp | High-Vol<br>(3)<br>FinComp | Low-Vol<br>(4)<br>FinComp | High-R&D<br>(5)<br>FinComp | Low-R&D<br>(6)<br>FinComp |
| Beta                                                        | 3.549***<br>(0.035)     | 3.005***<br>(0.048)     | 5.669***<br>(0.043)        | 7.827***<br>(0.050)       | 7.530***<br>(0.039)        | 6.862***<br>(0.044)       |
| Observations                                                | 7,500,000               | 7,500,000               | 7,500,000                  | 7,500,000                 | 7,500,000                  | 7,500,000                 |
| R-squared                                                   | 0.149                   | 0.137                   | 0.104                      | 0.127                     | 0.115                      | 0.093                     |
| Cluster Level                                               | Firm                    | Firm                    | Firm                       | Firm                      | Firm                       | Firm                      |
| Controls                                                    | Y                       | Y                       | Y                          | Y                         | Y                          | Y                         |
| Year FEs                                                    | Y                       | Y                       | Y                          | Y                         | Y                          | Y                         |

**Table 11**

Regressions of risky financial asset holdings—sample splits. This table reports—for the subsample splits by firm size (Panel A), volatility (Panel B), and R&D (Panel C)—the results of the regression in Eq. (26),  $FinAssets_{i,t} = \zeta_{i,t} + \phi \beta_{i,t}^{FD} + Control_{i,t} + \epsilon_{i,t}$ , where  $j$  and  $i$  index firm and year, respectively;  $FinAssets_{i,t}$  is either the fair value of risky financial assets over total assets (*Risky* in Columns 1, 2, 5 and 6), or the fair value of risky financial assets over total financial assets (*FinComp* in Columns 3, 4, 7 and 8);  $\zeta_{i,t}$  are 2-digit SIC code industry  $\times$  year fixed effects; the control variables are *LQ*, *LProf*, *LSize*, and *FDResVol*; and  $\beta_{i,t}^{FD}$  is the financing deficit beta, is estimated for each firm from the time-series regression in Eq. (25),  $FD_{i,t} = \alpha_i + \beta_i^D R_{i,t}^B + \epsilon_{i,t}$ , where  $FD_{i,t}$  is the financing deficit of firm  $i$  in quarter  $t$ ;  $R_{i,t}^B$  is either the return on the Bloomberg US Aggregate Bond Market Index ("Agg Bond") or the reverse change in the 10-year Treasury bond yield ("Treasury") during quarter  $t-1$ ; and  $FDResVol$  is the standard deviation of the financing deficit residuals,  $\epsilon_{i,t}$ , from the corresponding regressions. Firms are split by yearly median size in Panel A, median financing deficit volatility in Panel B, and positive and zero R&D in Panel C. Standard errors in parentheses are clustered at the industry level and adjusted using the procedure in Murphy and Topel (1985) to account for sampling error in the first stage. Significance levels at 10%, 5%, and 1% are denoted by \*, \*\*, and \*\*\*, respectively. Variable definitions are in Table A.4.

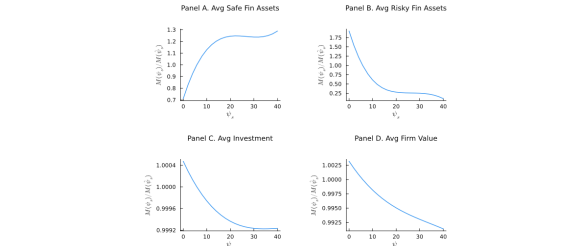
| Panel A. Split by Size |                     |                     |                     |                     |                  |                    |                  |                    |
|------------------------|---------------------|---------------------|---------------------|---------------------|------------------|--------------------|------------------|--------------------|
| VARIABLES              | Small Firms         |                     |                     |                     | Large Firms      |                    |                  |                    |
|                        | (1)<br>Risky        | (2)<br>Risky        | (3)<br>FinComp      | (4)<br>FinComp      | (5)<br>Risky     | (6)<br>Risky       | (7)<br>FinComp   | (8)<br>FinComp     |
| Beta                   | 0.310***<br>(0.057) | 0.079***<br>(0.014) | 0.544***<br>(0.107) | 0.144***<br>(0.030) | 0.091<br>(0.109) | 0.064**<br>(0.027) | 0.458<br>(0.319) | 0.197**<br>(0.076) |
| Observations           | 9652                | 9652                | 9590                | 9590                | 9646             | 9646               | 9540             | 9540               |
| R-squared              | 0.113               | 0.112               | 0.067               | 0.067               | 0.155            | 0.158              | 0.118            | 0.121              |
| Beta Type              | Agg Bond            | Treasury            | Agg Bond            | Treasury            | Agg Bond         | Treasury           | Agg Bond         | Treasury           |
| Cluster Level          | Industry            | Industry            | Industry            | Industry            | Industry         | Industry           | Industry         | Industry           |
| Controls               | Y                   | Y                   | Y                   | Y                   | Y                | Y                  | Y                | Y                  |
| Ind $\times$ Year FEs  | Y                   | Y                   | Y                   | Y                   | Y                | Y                  | Y                | Y                  |

| Panel B. Split by Volatility |                       |                     |                     |                     |                      |                  |                  |                   |
|------------------------------|-----------------------|---------------------|---------------------|---------------------|----------------------|------------------|------------------|-------------------|
| VARIABLES                    | High Volatility Firms |                     |                     |                     | Low Volatility Firms |                  |                  |                   |
|                              | (1)<br>Risky          | (2)<br>Risky        | (3)<br>FinComp      | (4)<br>FinComp      | (5)<br>Risky         | (6)<br>Risky     | (7)<br>FinComp   | (8)<br>FinComp    |
| Beta                         | 0.264***<br>(0.043)   | 0.068***<br>(0.014) | 0.566***<br>(0.131) | 0.159***<br>(0.038) | -0.202<br>(0.237)    | 0.095<br>(0.064) | 0.099<br>(0.643) | 0.314*<br>(0.157) |
| Observations                 | 9625                  | 9625                | 9518                | 9518                | 9643                 | 9643             | 9582             | 9582              |
| R-squared                    | 0.121                 | 0.121               | 0.080               | 0.081               | 0.100                | 0.101            | 0.089            | 0.090             |
| Beta Type                    | Agg Bond              | Treasury            | Agg Bond            | Treasury            | Agg Bond             | Treasury         | Agg Bond         | Treasury          |
| Cluster Level                | Industry              | Industry            | Industry            | Industry            | Industry             | Industry         | Industry         | Industry          |
| Controls                     | Y                     | Y                   | Y                   | Y                   | Y                    | Y                | Y                | Y                 |
| Ind $\times$ Year FEs        | Y                     | Y                   | Y                   | Y                   | Y                    | Y                | Y                | Y                 |

| Panel C. Split by R&D |                     |                     |                     |                     |                  |                     |                    |                     |
|-----------------------|---------------------|---------------------|---------------------|---------------------|------------------|---------------------|--------------------|---------------------|
| VARIABLES             | High R&D Firms      |                     |                     |                     | Low R&D Firms    |                     |                    |                     |
|                       | (1)<br>Risky        | (2)<br>Risky        | (3)<br>FinComp      | (4)<br>FinComp      | (5)<br>Risky     | (6)<br>Risky        | (7)<br>FinComp     | (8)<br>FinComp      |
| Beta                  | 0.283***<br>(0.045) | 0.064***<br>(0.016) | 0.444***<br>(0.132) | 0.135***<br>(0.037) | 0.150<br>(0.094) | 0.069***<br>(0.024) | 0.636**<br>(0.244) | 0.199***<br>(0.068) |
| Observations          | 9663                | 9663                | 9581                | 9581                | 9618             | 9618                | 9532               | 9532                |
| R-squared             | 0.115               | 0.114               | 0.075               | 0.075               | 0.082            | 0.087               | 0.091              | 0.092               |
| Beta Type             | Agg Bond            | Treasury            | Agg Bond            | Treasury            | Agg Bond         | Treasury            | Agg Bond           | Treasury            |
| Cluster Level         | Industry            | Industry            | Industry            | Industry            | Industry         | Industry            | Industry           | Industry            |
| Controls              | Y                   | Y                   | Y                   | Y                   | Y                | Y                   | Y                  | Y                   |
| Ind $\times$ Year FEs | Y                   | Y                   | Y                   | Y                   | Y                | Y                   | Y                  | Y                   |



**Fig. 8.** Trading costs, financial assets, and firm value. This figure reports comparative ratios with respect to the trading cost parameter  $w_t$ . All other parameters are held fixed at their estimated values from Table 1. Each panel plots the ratio  $M(w_t)/M(w_0)$ , where  $M(\cdot)$  denotes the corresponding simulated moment and  $w_t$  is the estimated value reported in Table 1. Thus, a value of one corresponds to the benchmark economy at  $w_t$ , and deviations from one indicate proportional changes in the corresponding moment, rather than levels or percentages. Panel A corresponds to the moment of average safe financial asset holdings, Panel B to average risky financial asset holdings, Panel C to average investment, and Panel D to average firm value.

**Table 12**

Risky financial assets and firm value. This table reports the results of a counterfactual experiment based on the model to evaluate the effects and value of the option to invest in risky financial assets. We compare the full model to a counterfactual version in which firms are restricted from holding risky financial assets. Both models are solved using the estimated parameters reported in Table 1. We simulate two samples of firms from the full and counterfactual models and compare their outcomes across operating performance, financing behavior, and firm value. Column 1 reports the mean differences in each outcome, scaled by average capital in the full model—except for issuance frequency and firm value, which are scaled by their respective means. Columns 2 to 5 report analogous differences for subsamples defined by business cycle conditions and firm size, using the same scaling relative to conditional means. Boom and bust periods are defined by positive and negative aggregate profitability shocks, respectively; below and above median size firms are defined relative to the yearly median of total assets.

|                                  | % Differences No Risky vs Full Model |                     |                     |                      |                      |
|----------------------------------|--------------------------------------|---------------------|---------------------|----------------------|----------------------|
|                                  | (1)<br>Full Sample                   | (2)<br>Boom Periods | (3)<br>Bust Periods | (4)<br>< Median Size | (5)<br>> Median Size |
| Capital                          | -0.1685                              | -0.1617             | -0.1233             | -0.2289              | -0.1501              |
| Investment                       | -0.0165                              | -0.0170             | 0.0150              | -0.0398              | -0.0093              |
| Profits                          | -0.0250                              | -0.0213             | -0.0210             | -0.0305              | -0.0233              |
| Financial Assets                 | -0.3566                              | -0.2282             | -0.3305             | -0.6704              | -0.2610              |
| Tax Costs on Financial Assets    | -0.0018                              | -0.0011             | -0.0016             | -0.0033              | -0.0013              |
| Trading Costs                    | -0.0008                              | -0.0006             | -0.0006             | -0.0016              | -0.0006              |
| Cash Flows from Financial Assets | -0.0110                              | -0.0385             | -0.0824             | -0.0207              | -0.0080              |
| Distributions                    | -0.0176                              | -0.0439             | -0.1156             | -0.0086              | -0.0203              |
| Issuance Amount                  | 0.0471                               | 0.0879              | 0.0008              | 0.0997               | 0.0311               |
| Issuance Costs                   | 0.0045                               | 0.0084              | 0.0001              | 0.0095               | 0.0030               |
| Issuance Frequency               | 4.6964                               | 1.2243              | 19.2459             | 5.0150               | 4.2484               |
| Firm Value                       | -0.3499                              | -0.3089             | -0.2143             | -0.4594              | -0.3008              |

Table 13

Risky financial assets, investment, and external financing. This table reports the results of the regression in Eq. (27),  $Y_{it} = \varphi_1 Risk_{j,t-1} + \varphi_2 Safe_{j,t-1} + Controls_{j,t-1} + \alpha_j + \epsilon_t + \epsilon_{jt}$ , where  $j$  and  $t$  index firm and year, respectively; the dependent variable is either the investment rate (*InvTot* in Panel A) or external financing (*ExtFin* in Panel B); *LRisky* and *LSafe* are lagged risky and safe financial assets; control variables are *LQ*, *LProf*, and *LSize*; and firm and year fixed effects are included. Column 1 reports results for the full sample. Columns 2 and 3 report results for small and large firms, defined by whether total assets in the previous year fall below or above the annual median, respectively. Columns 4 and 5 report results for high- and low-volatility firms, based on whether financing deficit volatility is above or below the median. Columns 6 and 7 report results for high- and low-R&D firms, defined by whether average R&D expenditure is positive or zero, respectively. Standard errors are clustered at the industry level defined by 2-digit SIC code. Significance levels at 10%, 5%, and 1% are denoted by \*, \*\*, and \*\*\*, respectively. Variable definitions are in Table A.4.

| Panel A. Investment |                     |                     |                     |                     |                     |                     |                     |
|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                     | Full<br>(1)         | Small<br>(2)        | Large<br>(3)        | High-Vol<br>(4)     | Low-Vol<br>(5)      | High-R&D<br>(6)     | Low-R&D<br>(7)      |
| VARIABLES           | InvTot              | InvTot              | InvTot              | InvTot              | InvTot              | InvTot              | InvTot              |
| LRisky              | 0.186***<br>(0.061) | 0.298***<br>(0.102) | 0.012<br>(0.049)    | 0.222***<br>(0.067) | 0.120*<br>(0.072)   | 0.240***<br>(0.078) | 0.085*<br>(0.046)   |
| LSafe               | 0.439***<br>(0.062) | 0.463***<br>(0.076) | 0.301***<br>(0.090) | 0.549***<br>(0.083) | 0.140***<br>(0.030) | 0.468***<br>(0.087) | 0.400***<br>(0.102) |
| Observations        | 16,946              | 8396                | 8428                | 8409                | 8537                | 8516                | 8430                |
| R-Squared           | 0.070               | 0.062               | 0.106               | 0.073               | 0.091               | 0.068               | 0.116               |
| Cluster Level       | Industry            | Industry            | Industry            | Industry            | Industry            | Industry            | Industry            |
| Controls            | Y                   | Y                   | Y                   | Y                   | Y                   | Y                   | Y                   |
| Firm FEs            | Y                   | Y                   | Y                   | Y                   | Y                   | Y                   | Y                   |
| Year FEs            | Y                   | Y                   | Y                   | Y                   | Y                   | Y                   | Y                   |

| Panel B. External Financing |                      |                      |                   |                      |                   |                      |                     |
|-----------------------------|----------------------|----------------------|-------------------|----------------------|-------------------|----------------------|---------------------|
|                             | Full<br>(1)          | Small<br>(2)         | Large<br>(3)      | High-Vol<br>(4)      | Low-Vol<br>(5)    | High-R&D<br>(6)      | Low-R&D<br>(7)      |
| VARIABLES                   | ExtFin               | ExtFin               | ExtFin            | ExtFin               | ExtFin            | ExtFin               | ExtFin              |
| LRisky                      | -0.123***<br>(0.027) | -0.189***<br>(0.039) | -0.004<br>(0.032) | -0.160***<br>(0.037) | -0.054<br>(0.043) | -0.156***<br>(0.038) | -0.063**<br>(0.030) |
| LSafe                       | -0.154***<br>(0.042) | -0.193***<br>(0.043) | -0.018<br>(0.055) | -0.202***<br>(0.048) | -0.033<br>(0.058) | -0.181***<br>(0.045) | -0.089<br>(0.074)   |
| Observations                | 16,959               | 8401                 | 8435              | 8416                 | 8543              | 8518                 | 8441                |
| R-Squared                   | 0.126                | 0.134                | 0.072             | 0.153                | 0.064             | 0.143                | 0.115               |
| Cluster Level               | Industry             | Industry             | Industry          | Industry             | Industry          | Industry             | Industry            |
| Controls                    | Y                    | Y                    | Y                 | Y                    | Y                 | Y                    | Y                   |
| Firm FEs                    | Y                    | Y                    | Y                 | Y                    | Y                 | Y                    | Y                   |
| Year FEs                    | Y                    | Y                    | Y                 | Y                    | Y                 | Y                    | Y                   |

## 5.9 Table 13: risky assets, investment, and external financing in the data

Read:

- Table 13 connects the savings-portfolio evidence back to real corporate policies. Lagged risky financial assets are positively associated with investment and negatively associated with external financing.
- In the full sample, the coefficient on lagged risky assets is **0.186** in the investment regression and **-0.123** in the external-financing regression.
- The effects are much stronger for the constrained groups emphasized throughout the paper: small firms, high-volatility firms, and R&D-intensive firms.
- Safe assets have the same signs in the regressions, which confirms the paper's view that safe and risky assets are substitute liquidity tools. But risky assets are especially relevant when hedge demand is high.

## 6 Short summary

- **Method.** The paper estimates a dynamic model in which firms allocate precautionary savings between safe and risky securities and uses the financing deficit beta to summarize hedge demand.
- **Identification.** Structural parameters are pinned down from moments on profitability, investment, distributions, and safe/risky holdings; the empirical hedge-demand test comes from cross-sectional variation in firm-level financing deficit betas.
- **Main empirical finding.** Firms with financing deficits that co-move more positively with bond returns hold more risky financial assets, especially when those betas are tied to medium-

and long-term interest-rate risk.

- **Economic conclusion.** Risky financial assets help firms reduce costly external financing and support investment, with the effect strongest for small, volatile, and R&D-intensive firms.

## 7 Conclusion

### 7.1 Core conclusion

- The paper shows that investment in risky financial assets can arise as a fully rational equilibrium response to financing frictions and aggregate risk.
- The central sufficient statistic is the financing deficit beta: firms invest more in risky financial assets when those assets pay off in states in which financing deficits are large.
- In the data, this hedging channel is economically important and strongest for firms for which external financing is likely to be most costly.

### 7.2 Economic intuition

- Safe assets move liquidity across time. Risky assets move liquidity across both *time* and *states of the world*.
- When risky-bond returns are high in the same states in which the firm's investment demand is high, risky financial assets improve the match between available funds and desired investment.
- That lowers the expected use of costly external financing, and this is why risky assets raise firm value in the model.

### 7.3 Takeaway

This paper is useful because it changes how to think about corporate financial asset portfolios. Risky financial assets are not necessarily “wasteful cash management” or evidence of managerial distortions. They can be part of optimal liquidity policy. For firms whose financing needs are especially exposed to macroeconomic and interest-rate shocks, holding risky bonds is a way to hedge funding risk, sustain investment, and reduce dependence on external finance.